# MZUZU UNIVERSITY

## FACULTY OF INFORMATION SCIENCE AND COMMUNICATIONS

## DEPARTMENT OF INFORMATION COMMUNICATION AND TECHNOLOGY

**BICT 1202: PROGRAMMING IN C**

**SUPPLEMENTARY EXAMINATION**

**DATE: MAY, 2018** **TIME ALLOWED: 3 HOURS**

**INSTRUCTIONS**

i. This paper contains **One** page. **Please check.**
ii. There are **Two**questions in this paper. **Answer ALL.**
iii. Marks for each question are indicated.
iv. Write your answers clearly.

## Question One

a) Probably the best way to start learning a programming language is by writing a program. Below is an example program, study all the lines of the programme and explain the programme line by line.

- */* my first program in C */*
- *#include <stdio.h>*
- *int main ()*
- *{*
- *printft<< "Hello World!";*
- *return 0;*
- *}*

**[12 Marks]**

b) Explain the meaning of an infinite loop and write a programme with this kind of a condition. **[7 Marks]**

c) Write code to distinguish between function definition and function declaration. **[6 Marks]**

d) What is an array? **[2 Marks]**

e) Identify errors if any in the following and correct the errors so identified.

i. char keg[]={'A','B','C','D'}; **[2 Marks]**

ii. floatchar[7]={2.12, 4.0, 3.2, 3.14}; **[2 Marks]**

iii. intnumbers[2]={10,12,9,5}; **[2 Marks]**

## Question Two

a) Explain the significance of the break statement in a case sequence. **[3 Marks]**

b) Describe an identifier. **[5 Marks]**

c) Explain the importance of declaring a variablewith a type. **[6 Marks]**

d) Explain the meaning of this statement: intkomweka = akaka[6]; **[4 Marks]**

e) Write a program to input cost price and selling price of a product and check if a profit or loss was made after selling commodities. **[6 Marks]**

f) Write a programme that passes arguments to a function called **Kusintha** by **value**. **[6 Marks]**

g) Write a program to input marks of five subjects Physics, Chemistry, Biology, Mathematics and Computer from the user. Pass the marks by value to a function called grading. Calculate average and grade according to given conditions and out the mark and grade from the function: **[8 Marks]**

- If average >= 90% : Grade A
- If average >= 80% : Grade B
- If average >= 70% : Grade C
- If average >= 60% : Grade D
- If average >= 40% : Grade E
- If average < 40% : Grade F

**END OF QUESTION PAPER**